MTH129 Introduction to AI for Mathematical Sciences (MS)

2025-2026 Academic Year
Semester 2

SMP AI Team

Multi-Variable Calculus and Probability

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Main Contents

- Vector Calculus: Gradient and Jacobian
- Probability
  - (Continuous) Random Variable and PDF
  - Expectation and Variance
- Python

2.1 Multi-Variable Calculus

Introduction

- **Partial Derivative**: For a function f (a scalar): $\mathbb{R}^n \mapsto \mathbb{R}$, $\mathbf{x} \mapsto f(\mathbf{x})$, $\mathbf{x} = [x_1, x_2, \cdots, x_n]^T \in \mathbb{R}^n$ (a column vector) of n variables $x_1, \cdots, x_n$, the partial derivatives is defined as

$$\frac{\partial f}{\partial x_1} = \lim_{h \to 0} \frac{f(x_1 + h, x_2, \cdots, x_n) - f(x_1, x_2, \cdots, x_n)}{h}$$

$$\vdots$$

$$\frac{\partial f}{\partial x_n} = \lim_{h \to 0} \frac{f(x_1, x_2, \cdots, x_n + h) - f(x_1, x_2, \cdots, x_n)}{h}$$

- collect them in the row vector

$$\frac{\mathrm{d}f}{\mathrm{d}\mathbf{x}} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \cdots & \frac{\partial f}{\partial x_n} \end{bmatrix}$$

- The row vector is called the **gradient** of f
- Python code: np.gradient for gradient or sp.diff(f, x) for $\frac{\partial f}{\partial x}$. We need to "import sympy as sp" in advance.

Rules for Operations among Functions

Two functions f: $\mathbb{R}^n \mapsto \mathbb{R}$, and g: $\mathbb{R}^n \mapsto \mathbb{R}$

- **Product rule**:

$$\frac{\mathrm{d}f(\mathbf{x})g(\mathbf{x})}{\mathrm{d}\mathbf{x}} = \frac{\mathrm{d}f(\mathbf{x})}{\mathrm{d}\mathbf{x}}g(\mathbf{x}) + f(\mathbf{x})\frac{\mathrm{d}g(\mathbf{x})}{\mathrm{d}\mathbf{x}}$$

- **Sum rule**:

$$\frac{\mathrm{d}[f(\mathbf{x}) + g(\mathbf{x})]}{\mathrm{d}\mathbf{x}} = \frac{\mathrm{d}f(\mathbf{x})}{\mathrm{d}\mathbf{x}} + \frac{\mathrm{d}g(\mathbf{x})}{\mathrm{d}\mathbf{x}}$$

-

Two functions f: $\mathbb{R}^n \mapsto \mathbb{R}$, and g: $\mathbb{R} \mapsto \mathbb{R}$

- **Chain rule**:

$$\frac{\mathrm{d}}{\mathrm{d}\mathbf{x}}(g \circ f)(\mathbf{x}) = \frac{\mathrm{d}}{\mathrm{d}\mathbf{x}}(g(f(\mathbf{x}))) = \frac{\mathrm{d}g}{\mathrm{d}f}\frac{\mathrm{d}f}{\mathrm{d}\mathbf{x}}$$

Example 1

Assume that $f(\mathbf{x}) = x_1^3 x_2 + x_1^2 x_2$, $\mathbf{x} = [x_1 \ x_2]^T$, find $\frac{\mathrm{d}f}{\mathrm{d}\mathbf{x}}$.
Solution: We have

$$\frac{\partial f}{\partial x_1} = 3x_1^2 x_2 + 2x_1 x_2, \text{ and } \frac{\partial f}{\partial x_2} = x_1^3 + x_1^2$$

Hence,

$$\frac{\mathrm{d}f}{\mathrm{d}\mathbf{x}} = \left[\ 3x_1^2 x_2 + 2x_1 x_2, \qquad x_1^3 + x_1^2 \ \right]$$

2.2 Vector Calculus

- For a vector-valued function $\mathbf{f}$: $\mathbb{R}^n \mapsto \mathbb{R}^m$ and a vector $\mathbf{x} = [x_1, x_2, \cdots, x_n]^T \in \mathbb{R}^n$ (a column vector) of n variables $x_1, \cdots, x_n$, the corresponding vector of function values is defined as

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} f_1 \\ \vdots \\ f_m \end{bmatrix} \in \mathbb{R}^m,$$

  where each f_i(a scalar): $\mathbb{R}^n \mapsto \mathbb{R}$.

- The partial derivative of a vector-valued function $\mathbf{f}$: $\mathbb{R}^n \mapsto \mathbb{R}^m$ with respect to $x_i \in \mathbb{R}$, $i = 1, \cdots, n$, is

$$\frac{\partial \mathbf{f}}{\partial x_i} = \begin{bmatrix} \frac{\partial f_1}{\partial x_i} \\ \frac{\partial f_2}{\partial x_i} \\ \vdots \\ \frac{\partial f_m}{\partial x_i} \end{bmatrix}$$

- The gradient of $f\colon \mathbb{R}^n \mapsto \mathbb{R}^m$ with respect to $\mathbf{x} \in \mathbb{R}^n$ by collecting these partial derivatives

$$\frac{\mathrm{d}\mathbf{f}}{\mathrm{d}\mathbf{x}} = \begin{bmatrix} \frac{\partial \mathbf{f}}{\partial x_1} & \cdots & \frac{\partial \mathbf{f}}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} = \mathbf{J}$$

- The collection of all first-order partial derivatives of a vector-valued function $\mathbf{f}\colon \mathbb{R}^n \mapsto \mathbb{R}^m$ is called the **Jacobian**.

Example 2

Given

$$\mathbf{f}(\mathbf{x}) = \mathbf{A}\mathbf{x}, \quad \mathbf{f}(\mathbf{x}) \in \mathbb{R}^M, \quad \mathbf{A} \in \mathbb{R}^{M\times N}, \quad \mathbf{x} \in \mathbb{R}^N$$

Find $\frac{\mathrm{d}f}{\mathrm{d}\mathbf{x}}$.

Solution: $f_i(\mathbf{x}) = \sum_{j=1}^{N} A_{ij}x_j \Rightarrow \frac{\partial f_i}{\partial x_j} = A_{ij}$

We then have

$$\frac{\mathrm{d}\mathbf{f}}{\mathrm{d}\mathbf{x}} = \begin{bmatrix} \frac{\partial \mathbf{f}}{\partial x_1} & \cdots & \frac{\partial \mathbf{f}}{\partial x_n} \end{bmatrix} = \begin{bmatrix} A_{11} & \cdots & A_{1N} \\ \vdots & \ddots & \vdots \\ A_{M1} & \cdots & A_{MN} \end{bmatrix} = \mathbf{A}.$$

Chain Rule

Define $\mathbf{h}(\mathbf{x}) = \mathbf{g}(\mathbf{f}(\mathbf{x}))$, where $\mathbf{x} \in \mathbb{R}^{n_1}$, $f\colon \mathbb{R}^{n_1} \mapsto \mathbb{R}^{n_2}$, and $g\colon \mathbb{R}^{n_2} \mapsto \mathbb{R}^{n_3}$

- **Chain rule**:

$$\frac{\mathrm{d}\mathbf{h}}{\mathrm{d}\mathbf{x}} = \frac{\mathrm{d}\mathbf{g}}{\mathrm{d}\mathbf{f}}\frac{\mathrm{d}\mathbf{f}}{\mathrm{d}\mathbf{x}} = \begin{bmatrix} \frac{\partial g_1(\mathbf{y})}{\partial y_1} & \cdots & \frac{\partial g_1(\mathbf{y})}{\partial y_{n_2}} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_{n_3}(\mathbf{y})}{\partial y_1} & \cdots & \frac{\partial g_{n_3}(\mathbf{y})}{\partial y_{n_2}} \end{bmatrix} \begin{bmatrix} \frac{\partial f_1(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial f_1(\mathbf{x})}{\partial x_{n_1}} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_{n_2}(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial f_{n_2}(\mathbf{y})}{\partial x_{n_1}} \end{bmatrix}$$

Example 3

Let $g(x_1, x_2) = x_1^2 + 2x_2$ and $x_1 = \sin t$ and $x_2 = \cos t$. Find $\frac{\mathrm{d}g}{\mathrm{d}t}$
Solution: We have

$$\frac{\mathrm{d}g}{\mathrm{d}\mathbf{x}} = \begin{bmatrix} 2x_1 & 2 \end{bmatrix}, \text{ and } \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix}$$

Hence,

$$\frac{\mathrm{d}g}{\mathrm{d}t} = [2x_1 \ 2] \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix} = 2\sin t \cos t - 2\sin t$$

Properties & Self-study task

Properties

- $\frac{\mathrm{d}\mathbf{x}^T\mathbf{a}}{\mathrm{d}\mathbf{x}} = \mathbf{a}^T$
- $\frac{\mathrm{d}\mathbf{a}^T\mathbf{x}}{\mathrm{d}\mathbf{x}} = \mathbf{a}^T$
- $\frac{\mathrm{d}\mathbf{x}^T\mathbf{A}\mathbf{x}}{\mathrm{d}\mathbf{x}} = \mathbf{x}^T(\mathbf{A} + \mathbf{A}^T)$

Self-study task

- Taylor expansion

Probability of an Event

An **event** is a set of possible outcomes.
If A is an event, then we denote the **probability of** A by $P(A)$. Probabilities must satisfy the following properties:

1. $0 \leq P(A) \leq 1$ for every event A.
2. If S is the set of all possible outcomes, called the **sample space**, then $P(S) = 1$.
3. If events A and B are **disjoint**, that is, they have no outcomes in common, then $P(A \cup B) = P(A) + P(B)$. Here event $A \cup B$ contains all outcomes in either A or B.

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If A^c denotes the **complement of event** A, that is, the set of all outcomes in the sample space S that are not in the event A, then $P(A^c) = 1 - P(A)$.

Definition: Random Variable

A rule that assigns a numerical value to the outcome of an experiment is called a **random variable**.
For example, our experiment might be to toss three fair coins. We could define the random variable X to be the number of flowers on the three tosses.
We can get the sample space is $\{X = 0, X = 1, X = 2, X = 3\}$. The following is the **probability distribution** of X.

x	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Expectation and Variance: Discrete Case

Definition **Expectation of a Random Variable**

If X is a random variable with probability distribution

x	x_1	x_2	$\cdots$	x_n
$P(X = x)$	p_1	p_2	$\cdots$	p_n

then the **expectation** of X, denoted $E(X)$, also called the **mean** of X and denoted μ, is

$$\mu = E(X) = x_1p_1 + x_2p_2 + \cdots + x_np_n = \sum_{i=1}^{n} x_ip_i$$

The variance of a random variable X is the expected value of the squared deviation from the expected value of X, $\mathrm{Var}(X) = E[(X - E(X))^2]$.

So, for the previous example, $E(X) = 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8} = 1.5$ and $\mathrm{Var}(X) = \frac{1}{8}(0-1.5)^2 + \frac{3}{8}(1-1.5)^2 + \frac{3}{8}(2-1.5)^2 + \frac{1}{8}(3-1.5)^2 = 0.75$.

Continuous Random Variable

If the set of possible values of a random variable X is finite, such as $\{x_1, x_2, \cdots, x_n\}$, or is infinite, but can be put in a list such as $\{x_1, x_2, \cdots\}$, then the random variable X is said to be **discrete**. If a random variable X can take on any value in some interval of real numbers then we say that X is a **continuous random variable**.

There are many situations where, theoretically at least, the outcome can be any real number in an interval: for example, waiting time for a stop light, mass of a plastic molded part, or lifetime of a battery.

Probability Density Function (PDF)

For a continuous random variable X, we must specify the **probability density function (PDF)**. A PDF for a random variable X that takes on values in the interval $[A, B]$ is a **function** satisfying

1. $f(x) \geq 0$
2. $\int_A^B f(x)dx = 1$
3. $P(a \leq X \leq b) = \int_a^b f(x)dx$ for all a, b ($a \leq b$) in the interval $[A, B]$

Remark. $P(X = a) = 0$ for all a.

Expected Value and Variance

Let $f(x)$, $x \in [a, b]$ be the PDF of a continuous random variable X. The **expected value** of X is

$$E(X) = \int_a^b xf(x)dx$$

and the variance is

$$\begin{aligned}
\mathrm{Var}(X) &= \int_a^b (x - E(X))^2 f(x)dx \\
&= \int_a^b x^2 f(x)dx - 2\int_a^b xE(X)f(x)dx + \int_a^b E(X)^2 f(x)dx \\
&= E(X^2) - 2E(X)\int_a^b xf(x)dx + E(X)^2\int_a^b f(x)dx = E(X^2) - E(X)^2
\end{aligned}$$

Cumulative Distribution Function (CDF)

The **Cumulative Distribution Function (CDF)** of a random variable X is the function $F(x) = P(X \leq x)$.
For a continuous random variable X that takes on values on the interval $[A, B]$ and having PDF $f(x)$, the CDF is equal to

$$F(x) = \int_A^x f(t)dt, \qquad A \leq x \leq B$$

Let X be a continuous random variable taking on values in the interval $[A, B]$ and having PDF $f(x)$ and CDF $F(x)$. Then

1. $F'(x) = f(x)$
2. $F(A) = 0$ and $F(B) = 1$
3. $P(a \leq X \leq b) = F(b) - F(a)$

Example 4

Suppose a random variable X has CDF

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{2x}{x+1} & \text{for } 0 \leq x \leq 1 \\ 1 & \text{for } x > 1 \end{cases}$$

Find (a) $P(X < 2)$, (b) $P(0.5 < X < 0.7)$ and (c) the PDF $f(x)$.
Answer. (a) $P(X < 2) = P(X \leq 2) = F(2) = 1$.
(b) $P(0.5 < X < 0.7) = P(X < 0.7) - P(X \leq 0.5) = P(X \leq 0.7) - P(X \leq 0.5) = F(0.7) - F(0.5) = \frac{8}{51} \approx 0.15686$.
(c) $f(x) = F'(x) = \frac{2}{(x+1)^2}$.

Example 5

The time in minutes that it takes a worker to complete a task is a random variable with PDF $f(x) = \frac{6}{125}x(5-x)$, $0 \le x \le 5$.
(a) What is the probability that it takes more than 3 minutes to complete the task?
(b) Find the expected value of the time to complete the task and the variance.
(c) Find the CDF $F(x)$.

Answer. (a) $P(X > 3) = P(X \ge 3) = \frac{6}{125}\int_3^5 x(5-x)dx = \frac{44}{125} = 0.352$.
(b) $E(X) = \int_0^5 xf(x)dx = \frac{6}{125}\int_0^5 x^2(5-x)dx = \frac{5}{2} = 2.5$ and $E(X^2) = \int_0^5 x^2 f(x)dx = \frac{6}{125}\int_0^5 x^3(5-x)dx = \frac{15}{2} = 7.5$. So, $\mathrm{Var}(X) = E(X^2) - E(X)^2 = \frac{5}{4}$.
(c) $\int_0^x f(t)dt = \frac{3}{25}x^2 - \frac{2}{125}x^3$.

So, CDF $F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{3}{25}x^2 - \frac{2}{125}x^3 & \text{for } 0 \le x \le 5 \\ 1 & \text{for } x > 5 \end{cases}$

Gaussian Distribution

- A real-valued continuous random variable X is said to be a **Gaussian (normal) random variable** with parameters μ and σ, shown as $X \sim N(\mu, \sigma^2)$, if its PDF is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}},$$

  for all $x \in \mathbb{R}$.
- When $\mu = 0$ and $\sigma = 1$, that is $X \sim N(0, 1)$, X is a **standard Gaussian (standard normal) random variable**. Then the PDF is $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ for all $x \in \mathbb{R}$.
- If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$.
- For $X \sim N(\mu, \sigma^2)$, $E(X) = \mu$ and $\mathrm{Var}(X) = \sigma^2$.
- By S = np.random.normal(μ, σ, **M**), we can generate **M** random samples from the normal distribution $N(\mu, \sigma^2)$.

Python

- Function
- Vector Calculus
- Derivatives
- Integrals

Function

```python
# def is keyword to define a new function;
# return is keyword to return a value from a function
def f1(x, y, z=0):
    return (x+z)/y

f1(1,2) #uses default argument for z, i.e., z=0
f1(1,2,3) #overrides the default value of z
f1(z=1,x=2,y=3)
```

0.5
2.0
1.0

Function

```
def f2(string1, string2, separator=' '):
    return string1+separator+string2

f2('ab','cd')
f2('ab', ' ', 'cd')
f2('ab', 'cd', '_')
```

'ab cd'
'abcd '
'ab_cd'

Piecewise Function (by numpy)

Define a piecewise function $f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 2x & \text{for } 0 \leq x \end{cases}$.

```
def f(x):
    x = np.asarray(x)   # change the data type of x from
                        # floating scalar to numpy array.
    # Define conditions and the corresponding formulae
    conditions = [  x < 0,  x >= 0 ]
    functions = [  lambda x: 0, lambda x: 2 * x    ]
    result = np.piecewise(x, conditions, functions)
    # Return the result as scalar, instead of array
    if np.isscalar(x):
        return result.item()
    return result
```

Definite Integral

In numpy, we can calculate definite integral by integrate.quad from scipy.

```
from scipy import integrate

result, error = integrate.quad(f, 0, 2)
print(f"int f from 0 to 2 = {result}")
```

int f from 0 to 2 = 4.0

Piecewise Function (by sympy)

For exact symbolic derivative and integral of a piecewise function, we need to define the function by sympy.

Define a piecewise function $f(x) = \begin{cases} x^2 & \text{for } x < 0 \\ \frac{x^2}{x+1} & \text{for } 0 \leq x \leq 3 \\ \frac{3x}{4} & \text{for } 3 < x \end{cases}$.

```python
import sympy as sp

# Define symbols
x = sp.symbols('x')

# Define function
f = sp.Piecewise(
    (x**2, x < 0),
    ((3*x) /4, x > 3),
    ((x**2) / (x + 1), True))
```

(Exact symbolic) Derivative and Integral of Functions

By the following code, we can get the indefinite integral F and the derivative $\mathrm{f_prime}$.

```
f_prime = sp.diff(f, x)

F = sp.integrate(f, x)
```

Partial Derivatives and Gradient

```python
import sympy as sp

x, y = sp.symbols('x y')

f = x**3 * y + x**2*y
dfdx = sp.diff(f, x)      # partial derivative of f wrt x
dfdy = sp.diff(f, y)      # partial derivative of f wrt y
gradient = [dfdx, dfdy]

print(dfdx)
print(dfdy)
print(gradient)
```

$3 * x * *2 * y + 2 * x * y$
$x * *3 + x * *2$
$[3 * x * *2 * y + 2 * x * y, x * *3 + x * *2]$

Evaluate Gradient at a point

```
dfdxv = dfdx.subs({x: 2, y: -2})
dfdyv = dfdy.subs({x: 2, y: -2})
gradientv = [dfdxv, dfdyv]

print(dfdxv)
print(dfdyv)
print([dfdxv, dfdyv])
```

-32
12
$[-32, 12]$

```
dfdxv1 = dfdx.subs({x: 1.9, y: -2.1})
dfdyv1 = dfdy.subs({x: 1.9, y: -2.1})
print([dfdxv1, dfdyv1])
```

$[-30.7230000000000, 10.4690000000000]$

Derivative

```python
from sympy import symbols, diff
x = symbols('x')
y = symbols('y')
f = x**3+y**2
diff(f, x) # first-order derivative
diff(f, x, 2)  # second-order derivative
diff(f, y)
diff(f, y).subs({x: 3, y: 1}) # first-order derivative evaluated at x=3 and y=1
```

$3x^2$
$6x$
$2y$
2

Derivative

Generate 1000 random samples from the normal distribution $N(1, 4)$ and return the mean, standard deviation and (the first) ten elements.

```python
import numpy as np

S = np.random.normal(1, 2, 1000)

print(len(S))
print(np.mean(S))
print(np.std(S))
print("Ten elements in sample S:")
print(S[:10])
```

Practices

- Verify Examples 1, 4 and 5 by Python!
- Generate 2000 random samples from the standard normal distribution and return the mean, standard deviation and (the first) ten elements.
- Show that $S = (y_1 - \beta_1 x_{1,1} - \beta_2 x_{1,2})^2 + (y_2 - \beta_1 x_{2,2} - \beta_2 x_{2,2})^2 = (\mathbf{y} - \mathbf{x}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{x}\boldsymbol{\beta})$, where $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_{1,1} & x_{1,2} \\ x_{2,1} & x_{2,2} \end{bmatrix}$, and $\boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$. Find $\frac{\mathrm{d}S}{\mathrm{d}\boldsymbol{\beta}}$.

Thank you.